

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are: H H T H H H T H H T. Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Answer:

Heads = 7, Total Tosses = 10

Formula: $\hat{p} = \text{Number of Heads} / \text{Total Tosses}$

$\hat{p} = 7/10 = 0.70$

Final Answer: 0.70

2. Out of 50 students, 42 passed an exam. Find the MLE of the probability that a student passes.

Answer:

Passed = 42, Total Students = 50

$\hat{p} = 42/50 = 0.84$

Final Answer: 0.84

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

Answer:

Defective = 8, Total Bulbs = 100

$\hat{p} = 8/100 = 0.08$

Final Answer: 0.08

4. A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Answer:

Occurrences of 6 = 18, Total Rolls = 60

$\hat{p} = 18/60 = 0.30$

Final Answer: 0.30

5. Given Input (x)=4, Weight (w)=2, Actual Output=10, $y=w \times x$. Find predicted output, error and updated weight.

Answer:

Predicted Output = $2 \times 4 = 8$

Error = $10 - 8 = 2$

Updated Weight = $2 - (0.01 \times 4) = 1.96$

Final Answers: Predicted Output = 8, Error = 2, Updated Weight = 1.96

Code of Conduct:

I Maniyar Mohammad Rehan(192425391)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Mohammad Rehan

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation